Niklas Viebig

Email | LinkedIn | Github

MSc Physics @ ETH Zürich | Scientific Machine Learning & HPC | Climate & Astrophysics Applications

EDUCATION

University of Oxford — ***United Kingdom***
Visiting Research Student (Master's Thesis) — *Sep 2025 – Mar 2026*
ETH Zürich — ***Switzerland***
Master of Science - MSc, Physics — *Sep 2024 – Mar 2026*
Bachelor of Science - BSc, Physics — *Sep 2020 – Mar 2024*

RESEARCH & PROJECTS (Selected)

Master's Thesis: Differentiable Atmospheric Modeling with SpeedyWeather.jl — **Oxford, United Kingdom**
Supervisors: Milan Klöwer (University of Oxford), Prof. Siddhartha Mishra (ETH Zurich) — *Sep 2025 – Mar 2026*

- Building a differentiable version of the SpeedyWeather.jl model using Enzyme.jl to enable gradient-based parameter optimization in atmospheric simulations.
- Comparing reverse-mode AD with gradient-free methods and exploring hybrid ML–physics modeling, including neural network corrections to physical parameterizations.

Course Project: High Performance Computing for Weather and Climate — **Zürich, Switzerland**
Optimized Matrix-free Smoothers for Variable-Viscosity Stokes Equations — *Jul 2025 – Sep 2025*

- Learned and applied HPC fundamentals for weather and climate models: single-core optimization, shared-memory parallelism (OpenMP), distributed-memory parallelism (MPI), and GPU acceleration (CuPy, hybrid nodes) by designing and benchmarking optimized matrix-free stencil kernels for multigrid smoothing in incompressible Stokes equations with variable viscosity.
- Conducted large-scale performance tests on Alps, Switzerland's largest supercomputer (270 PFLOPS, TOP500 #8, operated by CSCS), demonstrating that the RAS-Jacobi smoother sustained >90% weak-scaling efficiency up to 64 cores and achieved up to 3× speedup over the Jacobi baseline.

Winner, START Hack 2025, UNCCD & G20 Global Land Initiative Track — **St. Gallen, Switzerland**
Interactive Dashboard for Sustainable Land Management in the Sahel Zone, Africa — *Mar 2025*

- Leveraged alternative data (geospatial & population density) to synthesize actionable insights for sustainable land management
- Developed an interactive geospatial dashboard visualizing 20 years of land cover, vegetation productivity, rainfall, and population density trends in the Sahel region to support sustainable land management and conflict mitigation.

Course Project: Bayesian Statistical Methods and Data Analysis — **Zürich, Switzerland**
Bayesian Spatio-Temporal Modeling of Amazonian Wildfire Activity — *Sep 2024 – Feb 2025*

- Applied Bayesian hierarchical modeling to predict wildfire activity across 558 municipalities in the Amazon region over 12 years, integrating meteorological and land-use transition data, analyzing key environmental drivers of wildfire dynamics.

Research: ETH Zürich, Exoplanet and Habitability Group — **Zürich, Switzerland**
Simulating Coronagraphic Exoplanet Detection with JWST. — *Feb 2022 - Dec 2022*

- Developed a Python-based tool to simulate coronagraphic observations with JWST's MIRI instrument
- Enabled realistic modeling of direct exoplanet detection scenarios by integrating noise, optical configurations, and detector effects bridging a simulation gap for proposal preparation and instrument characterization.

WORK EXPERIENCE

Research Analyst Intern, Strategy& - Part of the PwC Network — **Berlin, Germany**
Mar 2024 - Aug 2024

- Conducted fundamental research to identify sector-specific KPIs and business model disruptions across 7 industries using Refinitiv, Capital IQ, PitchBook, Orbis, Statista, and EMIS.
- Led innovation with AI adoption, reducing research time by 40% via automated data extraction and report generation.
- Designed Strategy& Share, centralizing consulting best practices and research resources for improved knowledge accessibility.

SCHOLARSHIPS

Swiss-European Mobility Programme (SEMP) Scholarship, ***ETH Zurich & Swiss Confederation*** — *2025-2026*

- Awarded grant for Master's thesis focused on machine learning in climate modeling (SpeedyWeather.jl).

SFOU Scholarship, ***Scholarship Committee of the Swiss Friends of Oxford University*** — *2025*

- Awarded grant in recognition of academic merit and to support Master's thesis work at the University of Oxford.

SKILLS & INTERESTS

Languages: Python (NumPy, pandas, scikit-learn, PyTorch, JAX, TensorFlow), Julia, C++, SQL
Tools: Git, Jupyter, MPI/OpenMP, CUDA/CuPy
Skills: Machine Learning (deep learning, NLP, generative models, Bayesian inference), Data Science (ETL, feature engineering, visualization, statistical modeling), High-Performance Computing (distributed training, GPU acceleration, performance optimization), Scientific Computing (PDE solvers, automatic differentiation, climate & risk modeling)